

DL for ENG

LECTURE 8.2

Transient Heat

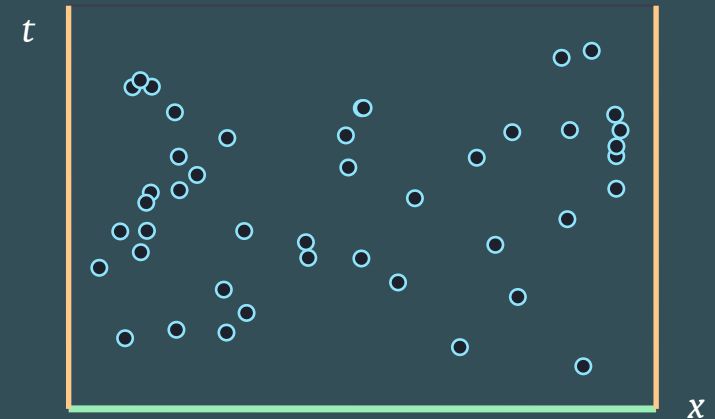
Remus Teodorescu

*With support from Research Assistant Noman Khan -
nomank@energy.aau.dk*

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

TIME · CAUSALITY · THE INVERSE PROBLEM



green: the field at $t = 0$ · amber: the walls

Time is one more input, and the whole block of space and time is trained at once. That buys a solution readable at any instant, and it costs the arrow of time: the loss does not know that early instants come first. The same plate then gives up its diffusivity to a few sensors.

Learning Objectives

SECTIONS, IN ORDER

- 1 **Storage and diffusion** — heat stored, and how fast a change spreads
- 2 **Time** — the transient equation, the space-time block, the start, causality
- 3 **Plate and inverse** — the plate in time, and α recovered from sensors

READ, AND WHAT EACH ANSWERS

Incropera ch. 5	how does a body store and spread heat?
Liu ch. 4.1 - 4.4	what does time add, and in what order?
Liu ch. 4.6	how is α found from sensors?

Heat Storage

Heat storage — a material that warms up takes in energy: the heat stored in each volume is ρc_p times its temperature rise. In steady state nothing is stored. In a transient, the heat that arrives and is not conducted away is stored, and the temperature changes.

ENERGY STORED PER VOLUME

$$e = \rho c_p (T - T_0)$$

e the energy stored in J/m³, ρ the density, c_p the specific heat in J/(kg·K)

- **steady** — nothing is stored: heat in equals heat out
- **transient** — the difference is stored, and T changes
- **examples** — a chip in a power pulse, a motor at start-up



HEAT CAPACITY PER VOLUME

material	ρc_p , MJ/(m ³ K)
copper	3.45
aluminium	2.43
silicon	1.64
stainless steel	4.00
glass	2.10
water	4.18
air	0.0012

Typical values at room temperature. Solids and water store about the same per volume, two to four MJ/(m³K); air stores three thousand times less.

Thermal Diffusivity

Thermal diffusivity — how fast a temperature change spreads through a material: the conductivity divided by the heat stored per volume, $\alpha = k/(\rho c_p)$. In a time t a change at the surface reaches a depth of about the square root of αt .

CONDUCT AGAINST STORE

$$\alpha = \frac{k}{\rho c_p}, \quad \delta \approx \sqrt{\alpha t}$$

α the diffusivity in m^2/s , δ the depth a change reaches in the time t

- **large** α — conducts well, stores little: fast
- **small** α — slow to respond: glass, water
- **example** — silicon: 0.30 mm in 1 ms, 9.57 mm in 1 s

A CHANGE AT THE SURFACE, SPREADING INWARDS



after t : the change has reached δ

after $4t$: only twice as deep

DIFFUSIVITY OF COMMON MATERIALS

material	α , mm^2/s
copper	116
aluminium	98
silicon	91
stainless steel	3.75
glass	0.48
water	0.15
air	22

Typical values: the conductivities of the last lecture over the heat capacities of the last slide. Air conducts badly but stores almost nothing, so a change spreads through it faster than through steel.

Transient Conduction

Transient conduction — the heat equation with units:

$\partial T / \partial t = \alpha \nabla^2 T + Q / (\rho c_p)$, where $\alpha = k / \rho c_p$ is the diffusivity. It is parabolic, so besides the walls it needs the field at $t = 0$. After the time constant τ a temperature difference is down to 37 %.

ONE NEW TERM, ONE NEW CONSTANT

$$\frac{\partial T}{\partial t} = \alpha (T_{xx} + T_{yy}) + \frac{Q}{\rho c_p}$$

$$\alpha = \frac{k}{\rho c_p} \quad [\text{m}^2/\text{s}]$$

$$\tau = \frac{L^2}{d \pi^2 \alpha}$$

τ the time constant, L the side of the plate, d the space dimensions: 2 for a plate
 α the diffusivity, k the conductivity, ρc_p the heat capacity per volume

THE TRANSIENT IN PYTORCH

```
xyt = torch.cat([x, y, t], 1)           # three inputs now
T = net(xyt)                             # the field in space and time
F = grad(T, t) - alpha * lap(T, xy) - Q_c # the residual; grad = autograd
tau = L**2 / (2 * pi**2 * alpha)         # the square's time constant
```

THE SLOWEST MODE DECAYS



A unit square with cold edges: its slowest mode, decaying and keeping its shape. Amber is τ , orange 4τ , where it has settled. At $t = 1$ the amplitude is 3×10^{-9} .

Space-Time Sampling

Space-time sampling — the interior points carry a third coordinate, t , and the whole block $\Omega \times (0, t_{\text{end}}]$ is trained at once. The solution is continuous in time and can be read at any instant, and nothing in the loss requires early times to be right before late ones.

A BLOCK, NOT A MARCH

$$(x, y, t) \in \Omega \times (0, t_{\text{end}}]$$

$$T(x, y, 0) = f_{IC}(x, y)$$

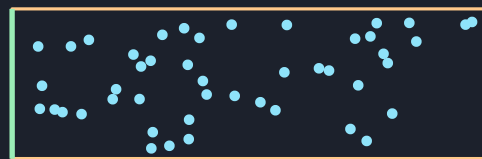
Ω the plate, t_{end} the end of the window, f_{IC} the initial field

- **no step size** — and no stability limit
- **any instant** — read the model, do not re-run it

A MARCH, AND A BLOCK



a solver marches, slice by slice



the network takes the whole block

$\Omega \times \text{TIME}$, SAMPLED AS ONE BLOCK



the slice

one condition, few points

the block

the residual, many points

Cyan is the residual, green the initial field: it is outvoted.

Initial Condition

Initial condition — the field at $t = 0$. As a loss term it is one slice among a block of interior points and is easily outvoted; built into the trial solution as $(1 - t) f_{IC} + t(\dots)\mathcal{N}$ it holds exactly, which assumes a time axis scaled to zero to one.

A TERM, OR A FACTOR

$$T = (1 - t)f_{IC}(x, y) + tx(1 - x)y(1 - y)\mathcal{N}_{\hat{w}}$$

f_{IC} the initial field, $\mathcal{N}_{\hat{w}}$ the network; at $t = 0$ the second term vanishes

- **soft** — three terms compete, and the slice loses
- **hard** — two terms disappear, and $t = 0$ is exact

TALK • 2 MIN How would you explain why the initial condition is the term a network loses most often, in your own words? Then, in the card on the right, what does the $(1 - t)$ factor assume, and what would break if t_{end} were ten?

THE TRIAL SOLUTION, TERM BY TERM

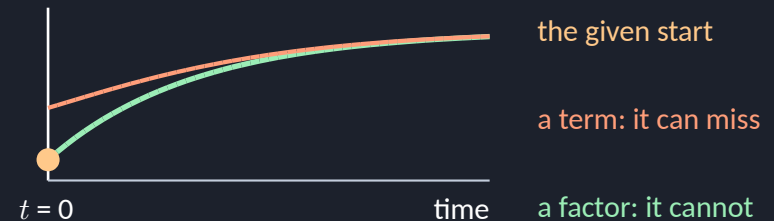
$(1 - t) f_{IC}$ the initial field, fading

$t \cdot x(1-x) y(1-y)$ zero at $t = 0$ and on the walls

$N(x, y, t)$ free everywhere else

Each factor switches off exactly one condition.

THE FIRST INSTANTS AT ONE POINT



Causality

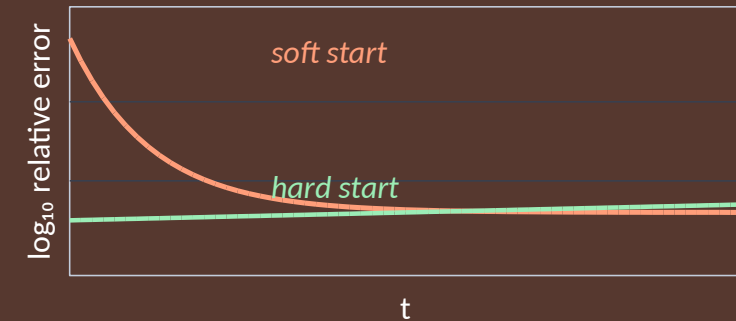
Causality — a solution at one instant follows from the earlier ones, but the loss sums over the block and treats $t = 0.01$ exactly as $t = 0.99$. A network can fit the late times first and carry an error at $t = 0$, the one instant that was known before training began.

THE ARROW OF TIME IS NOT IN THE LOSS

- **the loss** — one mean over the whole block
- **the symptom** — the largest error sits at $t = 0$
- **worse in reverse** — late times pull the early ones

TALK • 2 MIN How would you explain what causality means for a solution in time, in your own words? Then, in the plot on the right, why does the soft-start error grow towards $t = 0$, and would more interior points fix it?

SCHEMATIC • ERROR AGAINST TIME



FOUR REMEDIES, IN ORDER

hard start nothing to tune; Ex_08.2

causal weights effective; adds ϵ

time windows long transients; more code

early points cheap; Ex_08.2

The Plate in Time

Transient plate — the plate of L8.1 with a time coordinate: uniform generation, the hole held at $T = 0$, insulated edges and a cold start. The trial solution $t\phi\mathcal{N}$ holds the hole and the initial field at once, and the edges stay a loss term.

ONE FACTOR, TWO CONDITIONS

$$T = t\phi(x, y)\mathcal{N}_{\hat{w}}(x, y, t)$$

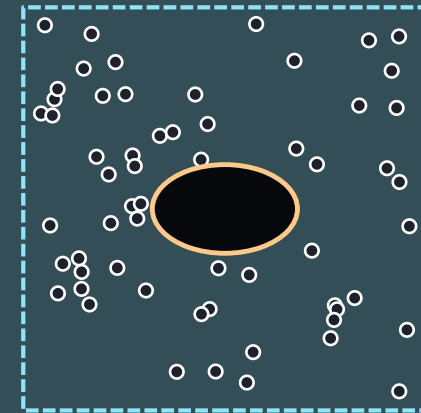
t holds the cold start, ϕ the level set holds the hole, $\mathcal{N}_{\hat{w}}$ the network

- **built in** — the hole, and the start at zero
- **a loss term** — the insulated edges, as before

THE PLATE IN PYTORCH

```
T = t * hole_multiplier(xy) * net(xyt)    # hole and cold start, exact
F = grad(T, t) - alpha * lap(T, xy) - Q_c # the transient residual
loss = F.pow(2).mean() + w_N * L_edges    # the edges stay a term
```

THE PLATE, NOW WITH A CLOCK



starts	cold, at $T = 0$ everywhere
loses heat	through the hole only
settles in	0.13: only the hole cools it

Inverse Diffusivity

Inverse diffusivity — the diffusivity α found from measurements. It is one more trainable number beside the weights, not an output of the network. It enters the loss through the residual, and it is written $\alpha = e^\beta$ so that it cannot go negative.

ONE RUN, TWO UNKNOWNNS

$$\alpha = e^\beta, \quad \mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_{data}$$

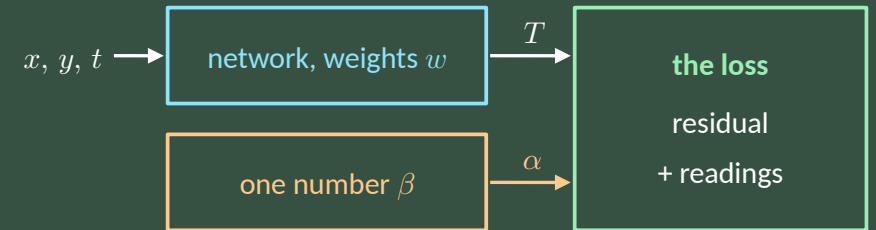
β the trained number, $\alpha = e^\beta$ the diffusivity, positive for any β
 $\mathcal{L}_{PDE}$ the residual, with α in it, $\mathcal{L}_{data}$ the misfit to the thermocouples

- **the data term** — what identifies α at all
- **the window** — the warm-up, while the readings change

THE INVERSE IN PYTORCH

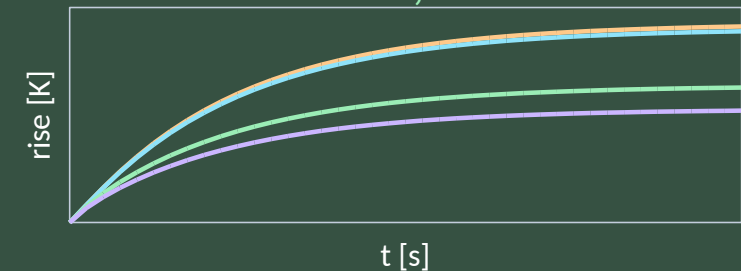
```
beta = nn.Parameter(torch.zeros(1))      # log  $\alpha$ , so  $\alpha$  stays positive
opt = torch.optim.Adam([*net.parameters(), beta]) # trained together
loss = F(net, beta.exp()).pow(2).mean() + data(T, x_s) # residual + data
```

ONE MORE TRAINABLE NUMBER



the optimiser changes w and β together

FOUR THERMOCOUPLES, WARMING UP



Ex_08.2: four thermocouples, every 2 s, noise 0.05 K. The warm-up carries α ; steady readings do not.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
lumped model	one temperature, when $Bi \ll 1$	Incropera ch. 5	a die in still air
causal weights	an instant weighted by its past	Liu ch. 4.4	a long transient
error and balance	relative and absolute, a balance	Liu ch. 4.6	no reference at hand
stiffness	scales that differ by decades	Liu ch. 4.10	thin films, fast pulses
other activations	sine and swish on a transient	Liu ch. 4.7.4	oscillating sources
walls that change	a flux that varies with time	Liu ch. 4.6.5	a switched source

WHY THEY ARE LEFT OUT

Ex_08.2 needs none of them: its plate has no single temperature, its start is built in, and it is scored against a finite-element reference. The others return with a switched source or the separated scales of L9.

Exercise 8.2

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the plate switched on: data, physics, scaled
- **4** — the reference built step by step; three meshes
- **5** — a PINN with the start built in; the three compared
- **6** — the diffusivity, from four thermocouples
- **7 – 9** — the report; mini projects MP8.2A and MP8.2B

THE QUESTION EACH PART ANSWERS

- 1 – 3** how long is the transient?
- 4** what is the answer, to 0.20 K?
- 5** how does a PINN compare?
- 6** can sensors give α ?

REPORT QUESTION, TO CONCLUDE

Finite differences gave the whole window in seconds. For which task in the notebook did the network do something the grid solver could not do directly?

Key Takeaways

TIME

what it is heat stored: one new term and input

function the field over a block of space-time

train, tune t_{end} from τ ; the start built in

vs steady an initial field, and no arrow of time

used for a device after a pulse, a cold start

PLATE AND INVERSE

what it is the plate of L8.1, started cold

function $t \neq N$: two conditions in one factor

train, tune against a reference; the warm-up

vs forward α trainable, found in the same run

used for properties you cannot measure

Next — L9.1: the same loop for a flow, where the nonlinearity is the difficulty.

Questions

1 - 2 • TIME

1 What does the heat equation add to steady conduction, and how does a PINN treat time?

A storage term, $\rho c_p \partial T / \partial t$, and so one initial field. The diffusivity $\alpha = k / \rho c_p$ says how fast a change spreads and sets the time constant τ . For a PINN time is one more input: the whole block of space and time is trained at once, and the solution can be read at any instant. *Liu ch. 4.1 - 4.3; Ex_08.2 sections 3 - 5*

2 What is causality, and how is it put back into training?

A solution at one instant follows from the earlier ones, but the loss treats all instants alike, so the network can fit late times first and carry an error at the start. Build the initial field into the trial solution and sample early times densely, as Ex_08.2 does; causal weights and a split window go further. *Liu ch. 4.2 - 4.4; Ex_08.2 section 5*

3 - 4 • PLATE AND INVERSE

3 How is the transient plate set up, and how is its solution checked?

The trial solution $t \varphi N$ holds the initial field and the hole wall at once; the insulated edges stay a loss term. With no exact solution, it is scored against a finite-element reference: the worst error over the plate at each instant, and the hottest point in time. *Liu ch. 4.6; Ex_08.2 sections 4 - 5*

4 How does an inverse problem differ from a forward one?

A parameter of the equation, here the diffusivity α , is trained beside the weights so that the solution fits measured data. It is optimised as $\log \alpha$, so it stays positive. Only the early part of the cooling curve constrains it: once the field has decayed, the data says nothing. *Liu ch. 4.6; Ex_08.2 section 6*

NEXT

L9.1 — Laminar Flow

The same loop again, on a velocity that carries itself: a nonlinear residual, and a pressure with no equation of its own.

BEFORE THE NEXT SESSION

read	Liu ch. 4.1 – 4.6
run	Ex_08.2, sections 1 to 7
bring	your error over time, and your α
think about	your own problem's time constant

Block eight is finished: the plate, steady and in time, forward and inverse.